

Vinod Suresh

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SUMMARY

Data & AI Program Lead with 9+ years of experience in Generative AI, LLM systems, MLOps, and cloud-scale ML platforms. Expert in RAG pipelines, AI observability, data governance, and Azure/AWS infrastructure. Proven record of translating complex AI systems into measurable business outcomes, including cost optimization, risk reduction, and compliance-aligned automation. Currently pursuing a Doctorate in AI & ML focused on Federated Learning.

TECHNICAL SKILLS

Programming Languages: Python, SQL, PHP
ML / AI: PyTorch, Transformers, YOLOv8, Isolation Forest, Federated Learning
Cloud & DevOps: Azure (AKS), AWS, Docker, Kubernetes
Data & MLOps: MLflow, Prometheus, Model Monitoring, Data Governance
Frameworks & APIs: FastAPI, Microservices Architecture
Databases & Graphs: MySQL, Neo4j
Analytics & BI: KPI Design, Dashboards, Time-Series Forecasting

Certifications

- AZ-900: Microsoft Azure Fundamentals
- AZ-305: Microsoft Azure Solutions Architect Expert
- AI-900 – Azure AI Fundamentals
- AI-102 – Azure AI Engineer Associate

EDUCATION

- Doctor of Business Administration (DBA) – Artificial Intelligence & Machine Learning Walsh College | 2024 – Present
- Post Graduate Certificate in Artificial Intelligence & Machine Learning McCombs School of Business | 2021 – 2022
- Bachelor of Computer Applications Indian University for Peace and Education | 2014 – 2017

WORK EXPERIENCE

CitiusTech Bangalore, India

Technical Lead

Oct 2024 - Present

- Designed LangChain-based RAG pipelines for healthcare AI agents, reducing hallucination rate by 40%.
- Architected microservice APIs integrating AI models on Azure AKS + FastAPI, achieving 99.8% uptime.
- Led AI governance and compliance initiatives, reducing deployment risk and accelerating enterprise AI approvals
- Partnered with analytics and compliance teams to establish unified data governance, improving audit readiness across multiple healthcare clients

MICROSOFT Bangalore, India

Sr. Machine Learning Engineer

Aug 2023 - Oct 2024

- Developed Prometheus + MLflow dashboards used across five internal Microsoft products, reducing false alerts by 30%.
- Engineered real-time AI threat monitoring tools, reducing incident response time and improving model reliability across enterprise systems
- Collaborated with legal, data privacy, and Responsible AI teams to ensure GDPR compliance for enterprise AI solutions.

- Defined reliability and factuality KPIs for production AI systems; standardized observability metrics across five internal products.

AB-INBEV Bangalore, India

Principal Analyst

Oct 2022 - May 2023

- Led econometric modeling of automation ROI in finance workflows, influencing \$10M+ in strategic investments.
- Delivered time-series forecasting dashboards supporting executive financial decisions and cost optimization.

PANTH.APP Bangalore, India

Backend Developer

Mar 2022 – Aug 2022

- Deployed anomaly detection pipelines using Isolation Forest for threat identification and alerting, demonstrating AI vulnerability analysis.
- Modeled system uptime patterns using Z-score and Isolation Forest algorithms.

Ederra Technologies & Vikramaditya Enterprises LLP - Bangalore

Software Engineer / Developer

June 2017 – Sept 2021

- Developed CRM + ERP tools powered by MySQL/PHP, automating 20% of sales cycle; achieved 97% client satisfaction.
- Created business dashboards and analytics systems adopted company-wide for performance tracking.

PROJECTS

SceneSenseAI – Multimodal Gameplay Narration Engine (*Paper Publication*)

- Designed multimodal pipeline (YOLOv8 + GPT-4) for real-time object, emotion, and mission-type prediction.
- Achieved 92% accuracy in gameplay narration, enabling automated content generation and reduced manual labeling effort.
- Filed patent and published (Scopus Indexed).
- Designed cross-functional data workflows integrating ML observability with BI dashboards for decision enablement.

Safety Metrics Evaluators For Risk-Optimized Ai

- Developed an adaptive driving assistant integrating perception, fusion, and decision layers for real-time risk and comfort optimization.
- Built an end-to-end AI pipeline leveraging contextual signals (weather, road, telemetry, driver intent) to predict optimal speed and cabin parameters.
- Trained Transformer-based Context Fusion model achieving 99% rule adherence and improved comfort stability in simulation.
- Implemented modular architecture (Python + PyTorch + FastAPI + Neo4j) with evaluators for safety metrics and performance visualization.

QoS–QoE Intelligence Framework for XR Systems (5G/6G Networks)

- Architected an end-to-end QoS→QoE intelligence framework for XR applications, integrating network telemetry (latency, jitter, throughput, packet loss), system performance metrics, and large-scale user feedback to enable experience-aware optimization.
- Designed scalable data pipelines combining packet-level network analysis and VR system metrics with statistical validation, establishing clear QoS–QoE relationships under varying network conditions.
- Developed and evaluated predictive ML models (Linear Regression, Random Forest, SVR) with feature engineering to support adaptive XR service optimization and informed decision-making.

RESEARCH PUBLICATIONS & PATENTS

- Doctoral Research (Ongoing):
 - Developing universal policy representations for cross-embodiment robotics, leveraging GNNs, Transformers, and reinforcement learning on RL Bench and Open X-Embodiment datasets.
- Augmented Reality Processing Apparatus For Personalized Shopping Experience.